SOFTWARE REQUIREMENTS

Chat & Messaging Module

Community Hub App

Academic Year 2025–2026

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1. Project Overview

The Community Hub is a collaborative application built by four student teams. Its goal is to connect people through a community feed, collaborative events, a chatting section and a matchmaking one.
Our team (TSO) owns the **Chat & Messaging** module, covering all real-time communication: direct messages, group conversations, and the social layer connecting users.
This document describes the **complete functional requirements** for this module. It covers all planned features and a section of **potential** future enhancements under consideration.

2. User Accounts

2.1 Home Page – Owner: Rares

- When someone opens the module while not logged in, they land on a home page that presents two options: Log In and Sign Up.
- Neither the conversations nor any other feature can be accessed until the user has logged in.

2.2 Log In – Owner: Rares

- On the Log In page, the user enters their username and password to get into their account.
- A "Forgot password?" option is available for anyone who needs to recover access to their account.
- Once logged in, the user can now access the module's features.

2.3 Sign Up – Owner: Rares

- To create a new account, the user fills in the following details: Username, Password, Email address, Phone number, and Birthday.

3. User Profiles

3.1 Viewing Your Own Profile – Owner: Eduard

- Users can view their own profile, which shows their username, profile picture, bio, current status, and birthday.
- Users can update their bio, profile picture, and birthday at any time.
- Users can switch their status between Online, Offline, and Busy whenever they like.

3.2 Viewing Other Users' Profiles – Owner: Eduard

- Users can open anyone else's profile to see their username, profile picture, bio, current status, and birthday.
- The profile shows how many mutual friends the two users share.
- From the profile, users can send a friend request or block that person.

3.3 Online Presence – Owner: Eduard

- Each user's status (Online, Offline, or Busy) is visible everywhere in the app: in the friends list, user list, group member panels, and their profiles.
- Status changes appear across the app in near real time.
- Users can choose to appear Offline even while actively using the app, effectively going “invisible”.

3.4 Birthday – Owner: Eduard

- On a user's birthday, a small birthday indicator appears next to their name in the friends list, user list, group member panels, and their profiles.

3.5 Mutual Friends – Owner: Eduard

- Interacting with the number of mutual friends shows the full list of shared friends.

4. Friends & Social Connections

4.1 Sending & Managing Friend Requests – Owner: Fineas

- Users can send a friend request to anyone from the user list.
- A dedicated tab shows all incoming friend requests, each with Accept and Decline options.
- Accepting a request automatically opens a direct message conversation between the two users.
- Declining a request removes it.

4.2 Friends List – Owner: Fineas

- Users can see all their current friends, with each entry showing their profile picture, username, and current status.
- Users can open a direct message with any friend directly from this list.

4.3 Blocking – Owner: Fineas

- Users can block someone from their profile, a direct message conversation, or the group member panel.
- Blocking prevents the blocked user from sending direct messages to the person who blocked them.
- Blocked users can still appear in shared groups but cannot interact privately with the person who blocked them.

5. Conversation List

5.1 Tabs – Owner: Alex

- The conversation list includes the following tabs: All, Direct Messages, Groups, Favorites, and Unread.
- The Unread tab shows every conversation with at least one unread message.
- The Favorites tab shows all conversations marked as favorites.

5.2 Conversation Preview – Owner: Alex

- Each entry shows the contact's name or avatar, a preview of the most recent message, the time it was sent, and an unread message count badge.
- Conversations with unread messages are visually highlighted.
- If the last message was deleted, the preview shows a message accordingly.

5.3 Favorites – Owner: Alex

- Users can mark any conversation as a favorite.

5.4 Search – Owner: Alex

- A search bar lets users filter conversations by Group name or by someone’s username (for DMs).

6. Direct Messages (DMs)

6.1 Opening & Creating DMs – Owner: Andrei

- A user can’t send messages to someone who is not in their friends list.
- A direct message conversation is created automatically when two users become friends, with a colored border that disappears once the conversation is opened for the first time.

6.2 DM Conversation View – Owner: Andrei

- Messages appear in chronological order, each showing the sender's profile picture, content, and timestamp.
- Edited messages display an "edited" label next to the timestamp.
- Deleted messages show "This message was deleted" in place of the original content.

6.3 Blocked DMs – Owner: Andrei

- If the user has been blocked by someone, the message input is disabled for both of them.

7. Group Conversations

7.1 Creating a Group – Owner: Cristi

- Users can create a group by giving it a name, optionally uploading a group icon (otherwise it's going to be a default icon), and selecting at least two other members.
- The person who creates the group automatically becomes the first admin.
- A notice specifying that the group has been created appears for everyone in it.

7.2 Member Panel (Right Side) – Owner: Cristi

- A collapsible panel on the right side of the screen lists all group members with their profile picture, username, current status, and role.
- Admins are clearly labelled within the panel.
- Admins can see when timeout expire for users.
- Admins can see what members are banned from the group.

7.3 Pinned Message – Owner: Cristi

- An admin can pin any message in the group. The pinned message then appears in a banner at the top of the conversation.
- A notice specifying that the message has been pinned appears for everyone in the group.

7.4 Role Management – Owner: Cristi

- Groups have three roles: admin, member, and banned.
- Admins can promote members to admin, demote admins back to member, apply timeouts, and ban members.
- A notice specifying that a role has been changed appears for everyone in the group.

7.5 Joining & Leaving – Owner: Cristi

- Any member can leave a group at any time.
- If the last admin leaves, the oldest remaining member is automatically promoted to admin.
- If the group becomes completely empty, it is deleted.

8. Messaging

8.1 Sending Messages – Owner: Andrei

- Users can type messages at any time and send them.
- Messages with no text cannot be sent.

8.2 Message Editing – Owner: Andrei

- The sender of a message can edit its text content after sending.
- The original text is no longer available once the message has been edited, only the updated version is visible.

8.3 Message Deletion – Owner: Andrei

- The sender of a message (or a group admin) can delete a message.

8.4 Message History & Scrollback – Owner: Andrei

- The full message history is available to all current participants.
- Users who join a group after it was created can still see all messages that were sent before they were members.
- Users can scroll up to load older messages.

9. Message Interactions

9.1 Replies – Owner: Cristi

- Users can reply to any message by selecting the reply option on it.
- The reply appears as a new message with a small quoted preview of the original shown just above it.
- The quoted preview includes the original sender's name and a shortened version of the content.
- Selecting the quoted preview scrolls the conversation back to the original message.

9.2 Reactions – Owner: Cristi

- Users can react to any message with an emoji.
- Reactions are shown as emoji counters below the message.
- Each user can only have one reaction per message and picking a different emoji replaces the previous one.
- Selecting an existing reaction counter either adds or removes the user's own reaction of that type.

9.3 Mention System – Owner: Cristi

- Users can mention another participant in a group by typing @ followed by their username.
- The mention appears highlighted in the sent message.

10. Moderation & Safety

10.1 Banning a Member – Owner: Eduard

- An admin can permanently ban a member from the group.
- A banned member cannot send messages or see any new messages posted after the ban.
- An admin can lift the ban at any time, restoring the member's access.

10.2 Timing Out a Member – Owner: Eduard

- An admin can give a member a temporary timeout, preventing them from sending messages until it expires.
- Available durations: 10 minutes, 1 hour, 24 hours, and 7 days, or a custom time.
- Admins can end a timeout early at any time.

11. Read Receipts

11.1 Read Receipts in DMs – Owner: Fineas

- In a direct message, the sender can see whether the other person has read their message.
- A "Seen" indicator appears below the last message the other user has opened.

11.2 Read Receipts in Groups – Owner: Fineas

- In a group conversation, each message shows a count of how many members have read it (e.g. "Read by 4").

12. Search

12.1 User Search – Owner: Rares

- When adding a member to a group, admins can search for users by username.
- People who are already in the group are excluded from the search results.

12.2 Message Search – Owner: Rares

- Within any conversation, there is a search button that opens a message search panel.
- The user can search for words or phrases within the conversation history.
- Matching messages are listed with context snippets. Selecting a result scrolls to that message.

13. Cross-Team Integrations

13.1 Community – Owner: Alex

- Users can send friend requests to anyone from the community section of the app.

13.2 Events – Owner: Alex

- Groups have the option for admins to create events.
- After an event is created, a notice appears in the chat.
- Events are exclusive to the groups that they were created in.

13.3 Matchmaking – Owner: Alex

- When the matchmaking system pairs two users together, a direct message conversation between them is created automatically if one doesn't already exist.
- A message appears in that conversation letting both users know they have been matched.

14. Possible Future Enhancements

The following features have been identified as possible helpful additions but are not included in the current scope due to time and complexity constraints. They are documented here for consideration in future development.

14.1 Per-Conversation Theme / Custom Colour – Owner: N/A

Each user could choose a personal accent colour for any conversation, DM or group, visible only on their own screen. This would apply to chat bubble colours, header accents, and similar UI elements. Other participants would not be affected. The colour could be reset to the default at any time.

14.2 Pinned Messages in DMs – Owner: N/A

Either participant in a direct message could pin a message so it appears in a persistent banner at the top of the conversation. Only one message could be pinned at a time, and pinning or unpinning would generate a small activity note in the chat.

14.3 Group Nicknames – Owner: N/A

Users could set a custom display name for themselves within a specific group. This nickname would be visible to all members of that group instead of the user's regular username. Users could clear their nickname at any time to revert back to their username.

14.4 Image Attachments – Owner: N/A

Users could attach an image to any message they send. The image would be displayed inline within the conversation for everyone to see.

14.5 Link Previews – Owner: N/A

When a message contains a URL, the app could automatically fetch a title and

brief description from the linked page and display them as a small preview card below the message. If the link can't be reached or has no useful information, no card would be shown.

14.6 Polls – Owner: N/A

Users could post a poll inside a conversation, a question with multiple answer options that other members can vote on. Results would update in real time as votes come in.

14.7 Admin Choice Pop-Up – Owner: N/A

If the last admin leaves, a pop-up appears to promote another member before leaving.

14.8 Unread Separator – Owner: N/A

When opening a conversation with unread messages, it should automatically scroll to the first unread message, showing a line as a separator between the messages that had been seen and the ones that are unread.

14.9 Read-by List – Owner: N/A

When the user selects the number of people who've read their message, a list containing those users should appear.